Java Hello World in Linux

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Time Required: 60 minutes

We are going to learn how to compile and run a Java program on Linux. You can follow the operational steps to compile a program from Java to bytecode.

This tutorial shows you how to compile a Java program on Kali Linux using an opensource JDK compiler. These directions can be used on any Debian flavored Linux.

NOTE: Commands, filenames, and directories are case sensitive.

Common Linux Commands

su – (change logon to root)

sudo (runs a command as root)

ifconfig

ls (lists files or directories)

cd (change directories)

mkdir (makes directories)

rmdir (removes directories)

cp (copy file)

rm (delete file)

chmod (modify file and directory security)

grep (search text file for characters)

passwd (create or change password)

Setup the Development Environment

Kali Linux is available in a prebuilt virtual machine.

1. If you do not have VirtualBox on your computer, go to www.virtualbox.org.
2. Download and install the version for your operating system.
3. Go to https://www.kali.org/get-kali/#kali-virtual-machines
4. Download the 64-bit VirtualBox file. This can take some time as the file is 3 GB.
5. Go to https://www.7-zip.org/download.html Download and install 7-zip for x64it Windows.
6. Right click the Kali Linux file you downloaded → 7-zip → Extract here.
7. Double Click the file you extracted. It will ask to be imported into VirtualBox.
8. Start Kali Linux.
9. The username and password is **kali**

The Java JRE (Java Runtime Envivornment) is already installed. We need to install the JDK to create programs.

10. Open a terminal. Type in the following.

```
sudo apt update
sudo apt install openjdk-17-jdk
```

11. Accept and install.

Let's confirm the installation.

```
java -version
```

Example run.

```
┌──(user㉿kalibill)-[~]
└─$ java --version
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
openjdk 17.0.5 2022-10-18
OpenJDK Runtime Environment (build 17.0.5+8-Debian-2)
OpenJDK 64-Bit Server VM (build 17.0.5+8-Debian-2, mixed mode, sharing)
```

Hello World

1. Create a directory and a sub directory to hold your Java programs and your main HelloWorld program.
2. Type **ls** to confirm. You should see a **Java** folder. Use the cd command to change directories.

```
┌──(user㉿kalibill)-[~]
└─$ mkdir Java

┌──(user㉿kalibill)-[~]
└─$ ls
Desktop    Downloads  Music     Public     Videos
Documents  Java       Pictures  Templates

┌──(user㉿kalibill)-[~]
└─$ cd Java
```

Write the Code

We will be writing the code in a command line editor, nano.

1. At the terminal: **nano HelloWorld.java**
2. Enter the following code.

```
user@kalibill: ~/Java
File  Actions  Edit  View  Help
  GNU nano 7.1                    HelloWorld.java
/*
 * Name: HelloWorld.java
 * Written by:
 * Written on:
 * Purpose: Traditional Hello World in Java
 */

public class HelloWorld {
   public static void main(String[] args) {
      // Print a string literal to the console
      System.out.println("Hello World!");
      System.out.println("This is my first Java Application");
   }
}
```

3. Press **CTRL S** to Save the file.
4. Press **CTRL X** to **Exit**.
5. The Linux cat command is used to take a quick look at a text file. Type **cat HelloWorld.java** to confirm the file.

```
┌──(user㉿kalibill)-[~/Code]
└─$ cat HelloWorld.java
/*
 * Name: HelloWorld.java
 * Written by:
 * Written on:
 * Purpose: Traditional Hello World in Java
 */

public class HelloWorld {
   public static void main(String[] args) {
      // Print a string literal to the console
      System.out.println("Hello World!");
      System.out.println("This is my first Java Application");
   }
}
```

Compile and Run the Program

1. To compile the program: run the following command. If there are any errors, check your program.

```
┌──(user㉿kalibill)-[~/Java]
└─$ javac HelloWorld.java
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
```

2. To execute your program: run the following command.

```
┌──(user㉿kalibill)-[~/Java]
└─$ java HelloWorld
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
Hello World!
This is my first Java Application
```

Congratulations! You created your first Java program in Linux!

Assignment Submission

Attach screenshot files of the following in the Blackboard assignment submission.

1. A screenshot of your code in nano.
2. A screenshot of your program running.